

Network reconstruction approach conditions ecological inference in food web reconstruction

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Abstract: Aim

Ecological networks are widely used to assess community structure, stability, and responses to disturbance. Such networks often require model-based reconstructions (*e.g.*, based on traits or theoretical constraints); however, the extent to which these frameworks influence ecological inference remains unexplored. Here, we assess whether macroecological inference derived from ecological networks is robust to variation in reconstruction framework.

Location

Cleveland Basin, United Kingdom.

Time period

Toarcian extinction event (Early Jurassic, late Pliensbachian–late Toarcian, ~183 Ma).

Major taxa studied

Marine macrofossils.

Methods

We reconstructed four successive assemblages from an identical species pool using six contrasting food web reconstruction approaches spanning feasible (trait-based), realised (allometric and energetic), and structural (topological) network representations. For each community and reconstruction approach, 100 replicate networks were generated. We quantified macro-, meso-, and micro-scale network properties and assessed differences among reconstruction approaches using multivariate analyses. Pairwise interaction turnover was measured using link-based beta diversity. We then simulated species loss under multiple disturbance scenarios, allowing cascading extinctions, and compared predicted community states using mean absolute differences and rank concordance metrics between reconstruction approaches.

Results

Reconstruction framework strongly influenced inferred network topology, generating distinct structural signatures independent of species composition. Reconstruction approaches that were similar in network metrics often diverged in species-level interactions, with high -turnover among inferred link sets. During extinction simulations, scenario rankings were broadly consistent at the network level, but interaction-level outcomes and cascade dynamics varied substantially.

Main conclusions

Network reconstruction functions as a structural prior that conditions ecological inference. While some aggregate patterns are robust across reconstruction approaches, detailed interaction-level dynamics are highly contingent on reconstruction approach. Comparative network studies across

spatial or environmental gradients should therefore align reconstruction framework with inferential goals and explicitly evaluate sensitivity to reconstruction assumptions.

Keywords: Ecological networks, Biotic interactions, Community assembly, Environmental gradients, Interaction turnover, Trophic organisation, Ecosystem resilience, Macroecology

1 Introduction

Understanding how biological communities are organised and how species interact with each other is a central goal of ecology. While early efforts focused primarily on species richness and composition, there is growing recognition that ecological communities are structured not only by which species occur, but by how they interact (Thuiller *et al.*, 2024). Interaction networks are increasingly treated as macroecological state variables where they are used to compare community organisation across environmental gradients, to quantify β -diversity in interaction structure, to evaluate stability–complexity relationships, and to infer vulnerability under global change (Poisot *et al.*, 2015; Trøjelsgaard & Olesen, 2016; Tylianakis & Morris, 2017; Gravel *et al.*, 2019).

As a result, ecological networks now play a central role in comparative analyses spanning latitudinal gradients, disturbance regimes, and deep-time environmental transitions (Roopnarine, 2006; Poisot & Gravel, 2014; Michalska-Smith & Allesina, 2019; Dunhill *et al.*, 2024; Hao *et al.*, 2025). Implicit in this expansion is the critical assumption that network properties estimated across systems using various models are structurally comparable, and that differences among network properties reflect ecological signal rather than methodological artefact (Fründ *et al.*, 2016; Jordano, 2016). However, most ecological networks are not fully observed as interaction data are incomplete and sampling is uneven across historical and biogeographic contexts, across both present day and deep-time (Poisot *et al.*, 2021; Sandra *et al.*, 2025).

Interactions must often be inferred indirectly from traits (*e.g.*, body size), phylogeny, co-occurrence, or theoretical constraints (Morales-Castilla *et al.*, 2015; Strydom *et al.*, 2021). Network construction therefore constitutes a model-based inference step rather than a purely observational exercise. Different reconstruction frameworks encode distinct ecological assumptions about how interactions arise - whether as biologically feasible combinations of traits, energetically optimised realised diets, or topological structures constrained by macroecological regularities. These assumptions act as structural priors over network architecture (Petchey *et al.*, 2011; Guimarães, 2020; Gauzens *et al.*, 2025; Strydom *et al.*, 2026). If alternative reconstruction approaches systematically generate different trophic configurations, then comparative analyses risk conflating ecological differences among communities with artefacts introduced by reconstruction choice. The reliability of macroecological inference therefore depends not only on ecological data, but on the structural assumptions embedded in network reconstruction approaches.

34 Despite rapid methodological development in interaction inference, few studies have di-
35 rectly evaluated how alternative reconstruction frameworks influence macroecological
36 conclusions when applied to the same species pool. This gap is particularly consequential
37 for comparative research, where network metrics are routinely interpreted as indicators
38 of environmental filtering, disturbance intensity, evolutionary history, or community sta-
39 bility (Allesina & Tang, 2012; Poisot *et al.*, 2015; Delmas *et al.*, 2018). If reconstruction
40 approaches encode distinct structural constraints over interaction topology, then differ-
41 ences among communities may reflect reconstruction assumptions rather than ecological
42 processes.

43 Deep-time ecosystems provide an especially useful test of this issue because trophic
44 interactions are not observed directly and must be inferred from traits, co-occurrence, and
45 theoretical constraints (Roopnarine, 2006; Dunne *et al.*, 2008; Dunne *et al.*, 2014; Dunhill
46 *et al.*, 2024; Karapınar *et al.*, 2026). As a result, ecological networks in these systems are
47 inherently reconstruction-dependent, rendering the assumptions embedded within different
48 reconstruction frameworks both explicit and consequential. Among such systems, the
49 Early Toarcian Extinction Event (ETEE; ~183 Ma) provides a particularly informative
50 case study. The ETEE was a major Early Jurassic biotic crisis associated with rapid
51 climatic warming, widespread marine oxygen depletion, and substantial ecological turnover
52 (Kemp *et al.*, 2024). Although considered a second-order extinction event globally, it
53 nevertheless resulted in the loss of approximately 26% of marine genera worldwide (Little
54 & Benton, 1995). Ecological disruption was especially severe in shallow marine ecosystems,
55 where environmental stress drove extensive restructuring of benthic communities and food
56 webs. In the Cleveland Basin of Yorkshire, the ETEE caused the extinction of around
57 60% of marine species, including up to 87% of benthic taxa (Caswell *et al.*, 2009). Fossil
58 assemblages from this interval therefore record both the collapse of established communities
59 and the subsequent recovery and reassembly of marine ecosystems under rapidly changing
60 environmental conditions. Recent work has shown that the event selectively impacted
61 specialist and infaunal organisms, leading to simplified ecosystems increasingly dominated
62 by ecological generalists and opportunistic taxa (Dunhill *et al.*, 2024). These changes were
63 accompanied by major shifts in community structure, trophic interactions, and ecosystem
64 connectivity, with evidence suggesting that ecological recovery following the extinction
65 was both prolonged and uneven across marine habitats.

66 Ecological networks have previously been used to infer patterns of community organisation

and extinction dynamics across this transition, providing an ideal test case for evaluating the robustness of network-based inferences of community response. In particular, recent work by Dunhill *et al.* (2024) has used reconstructed food webs to draw ecological conclusions about community stability and collapse during this event. This creates a rare opportunity to ask a more general question: to what extent are such inferences contingent on the specific reconstruction framework used to generate the network?

Here, we reconstruct ecological networks for four successive assemblages across the ETEE using six contrasting reconstruction frameworks spanning feasibility-based, realised, and structural models. By holding species composition constant while varying only the reconstruction approach, we isolate the influence of structural assumptions on inferred food web organisation, interaction turnover, and extinction dynamics. This design allows us to explicitly evaluate whether conclusions drawn from a single reconstruction approach (such as those of Dunhill *et al.* (2024)) are robust to alternative, equally plausible representations of interaction structure, or whether different reconstruction choices would lead to fundamentally different ecological inferences. More broadly, our study tests whether macroecological conclusions derived from ecological networks reflect underlying biological processes or are conditioned by the assumptions embedded within the reconstruction framework itself.

2 Methods

2.1 Study system and fossil data

We used fossil occurrence data from the Cleveland Basin, UK, spanning the late Pliensbachian to the late Toarcian. This interval encompasses a major volcanic-driven hyperthermal and marine extinction event. To capture network dynamics across this transition, we defined four successive palaeo-assemblages: pre-extinction (Pliensbachian), post-extinction (early Toarcian), early recovery (early–middle Toarcian) and late recovery (late Toarcian). Each taxon was characterized using their size and Bambach’s ecospace framework (Bambach *et al.*, 2007), coding for tiering, motility, and feeding mode as per Dunhill *et al.* (2024). Each assemblage was treated as a community of potentially interacting taxa. The dataset includes 57 taxa across diverse groups (*e.g.*, cephalopods, bivalves, and gastropods).

2.2 Network reconstruction approaches

2.2.1 Conceptual classification of network types

Ecological network reconstruction encompasses a range of approaches that differ in how trophic interactions are inferred and in the ecological assumptions they encode. These approaches can be broadly grouped into three classes: feasibility-based, realised, and structural models. Together, these classes represent distinct hypotheses about how interactions arise and constrain the space of possible food web structure.

Feasibility-based models infer the set of potential interactions among species based on trait compatibility or mechanistic rules. These approaches define a feasible interaction space by identifying which consumer–resource pairs are biologically possible, without specifying which interactions are realised in a given community. In palaeoecological contexts, where direct observation of interactions is not possible, such models provide a biologically grounded, formalised representation, of ‘expert knowledge’ based rules that can be used to infer interactions (*e.g.*, Fricke *et al.* (2022); Roopnarine (2006); Shaw *et al.* (2024)).

Realised models aim to approximate the subset of interactions that are expected to occur in practice. These approaches impose additional constraints (such as energetic optimisation, mechanical limits, or probabilistic niche structure) on top of feasibility, thereby generating networks that represent putative realised diets (Brose *et al.*, 2006; Schneider *et al.*, 2016). Although often parameterised using body size or related traits, realised models differ in their underlying ecological assumptions about how consumers select resources and how trophic interactions are structured.

Structural models, in contrast, do not attempt to reconstruct empirical trophic interactions from species-level data. Instead, they generate networks based on general topological constraints such as species richness, connectance, or trophic ordering (Allesina *et al.*, 2008). These models are species-agnostic and therefore cannot be interpreted as reconstructions of specific ecological communities. Rather, they have the potential to serve as ‘null hypotheses’, providing reference expectations against which the structure and dynamics of data-informed reconstructions can be evaluated.

In this study, we selected models to span these three conceptual classes (Table 1) and, critically, to sample distinct mechanistic interpretations of trophic interactions rather than exhaustively include all available implementations. Specifically, we include: a feasibility-

129 based model (PFWIM), representing trait-constrained interactions; multiple realised
130 models (ADBM, ATN, and Body-size ratio), capturing energetic optimisation, mechanical
131 constraints, and probabilistic allometric structure, respectively; and structural models
132 (Random and Niche), providing theoretical reference points. Although some approaches
133 share common inputs (*e.g.*, body mass), they encode fundamentally different assumptions
134 about how interactions arise. Conversely, some alternative palaeoecological approaches
135 (*e.g.*, guild- or feasibility-based frameworks) are conceptually aligned with PFWIM and
136 would therefore provide limited additional contrast for the purposes of this analysis.

Table 1: Six different network reconstruction approaches that can be used to construct food webs for both this specific community but are also broadly suited to palaeo network prediction. These reconstruction approaches span all facets of the network representation space (feasibility, realised, and structural network) and are suitable for an array of different palaeo-assemblages as the data requirements fall within the limitations set by the fossil record.

Recon- struction Approach	Assump- tions	Data needs	Limita- tion	Network type	Key reference	Usage examples
Random	Links assigned randomly	Species richness, number of links	Parameter assump- tions, species agnostic	Structural	Erdős & Rényi (1959)	Null- model compar- isons; testing whether observed network structure (con- nectance, motifs) deviates from random expecta- tions

Recon- struction Approach	Assump- tions	Data needs	Limita- tion	Network type	Key reference	Usage examples
Niche	Species ordered along a ‘niche axis’; in- teractions interval- constrained	Species richness, con- nectance	Parameter assump- tions, species agnostic	Structural	Williams & Martinez (2008)	Evaluat- ing trophic hierarchy and motif structure; baseline structural predic- tions
Allometric diet breadth model (ADBM)	Energy- maximizing predator diets	Body mass, abun- dance/biomass	Assumes optimal foraging; does not account for forbidden links	Realised	Petchey <i>et al.</i> (2008)	Predicting realized predator diets; exploring secondary extinc- tions
Allometric trophic network (ATN)	Links con- strained by body-size ratios and functional response	Body mass, number of basal species	Assumes only mechani- cal/ener- getic con- straints	Realised	Brose <i>et al.</i> (2006); Gauzens <i>et al.</i> (2023)	Simulat- ing species loss; evaluating network collapse dynamics

Recon- struction Approach	Assump- tions	Data needs	Limita- tion	Network type	Key reference	Usage examples
Paleo food web inference model (PFWIM)	Interac- tions inferred using trait- based mechanis- tic rules	Feeding traits	Assumes feeding mecha- nisms; trait resolution required	Feasibility	Shaw <i>et al.</i> (2024)	Mapping feasible trophic in- teractions; assessing secondary extinc- tions
Body-size ratio model	Probabilis- tic assign- ment of links based on predator- prey size ratios	Body mass	Does not account for forbidden links	Realised	Rohr <i>et al.</i> (2010)	Estimat- ing likely interac- tions; simulating cascading effects.

This classification allows us to distinguish variation arising from ecological assumptions embedded in reconstruction frameworks from variation attributable to species composition. By comparing networks generated from the same species pool across these complementary model classes, we explicitly evaluate how different representations of interaction structure condition ecological inference.

2.2.2 Network generation and replication

We evaluated six network reconstruction frameworks using the approaches listed in Table 1: Random and Niche models (structural networks); allometric diet breadth (ADBM), allometric trophic network (ATN), and Body-size ratio models (realised networks); and a paleo food web inference model (PFWIM; feasibility network). Expanded descriptions of reconstruction approach assumptions, parameterisation, and link-generation rules are provided in Supplementary Material S1, including full mathematical formulations and

parameter definitions. For each community, 100 stochastic network realisations were generated per reconstruction approach ($n = 2400$ per time bin in total, 600 per time bin) to capture uncertainty in link assignment. Where reconstruction approaches required species body mass or trait values, these were uniformly sampled within the different size classes, we adopted a uniform sampling by default, as alternative distributions (lognormal, truncated lognormal) have negligible impact on topology (Supplementary Material S1; Figure S1). Random and Niche models were parameterised using connectance values drawn from an empirically realistic range (0.05–0.25; Curtsdotter *et al.* (2011)). For the Random model, which requires link number explicitly, connectance (Co) was first converted to expected link number (L) using $L = Co \times S^2$, ensuring consistency in network density across all reconstruction approaches. Richness (S) was determined by the number of species for each time bin. For the Body-size ratio model, we followed the approach of Yeakel *et al.* (2014) and excluded the latent trait term, using only the body-mass scaling to determine links between species. Sensitivity of network structure to these parameter choices is evaluated in Supplementary Material S1. The PFWIM generates a deterministic feasible metaweb for a given trait configuration. To allow for statistical analyses we introduced controlled variation in interaction subsets by applying a downsampling procedure that stochastically samples feasible interactions. This step allowed us to introduce statistical variation while still preserving most of the metaweb structure. The choice of downsampling parameter and its effects on network structure are evaluated in Supplementary Material S1.

2.3 Network metrics and structural analyses

We quantified network structure using a suite of macro-, meso-, and micro-scale metrics Table 2, capturing overall network properties, motif structure, and species-level variability. Differences among reconstruction approaches were assessed using a multivariate analysis of variance (MANOVA), with reconstruction approach identity as a fixed factor and the full set of network metrics as response variables. Variance partitioning was further assessed using permutational multivariate analysis of variance (PERMANOVA). Pairwise interaction turnover was quantified using link-based β -diversity and was calculated following the framework of Poisot *et al.* (2012) among the four of the reconstruction frameworks (ADBM, ATN, body-size ratio, and PFWIM). Random and Niche models were excluded because they are species-agnostic reconstruction approaches that generate networks matching broad topological properties (*e.g.*, connectance and degree distributions) rather than

182 predicting biologically realistic species-specific interactions. Specifically, we looked at
183 interaction rewiring among shared species (β_{OS} , see S1 for additional details on the
184 decomposition of interaction -diversity and its implementation), which allows separation
185 of differences arising from altered interaction identities among species common to both
186 networks. Because all networks within a given assemblage are constructed from the same
187 species pool, differences in interaction structure primarily reflect ‘rewiring’ of trophic
188 links due to stochasticity in the reconstruction approach rather than species turnover.
189 All calculations were performed for all reconstruction combinations within the same
190 assemblage (time bin).

Table 2: Network properties used for analysis.

Metric	Definition	Scale	Reference
Connectance	L/S^2 , where S is the number of species and L the number of links	Macro	
Maximum trophic level	Prey-weighted trophic level averaged across taxa	Macro	Williams & Martinez (2004)
No. of linear chains (S1)	Number of linear chains, normalised by L/S	Meso	Milo <i>et al.</i> (2002); Stouffer <i>et al.</i> (2007)
No. of omnivory motifs (S2)	Number of omnivory motifs, normalised by L/S	Meso	Milo <i>et al.</i> (2002); Stouffer <i>et al.</i> (2007)
No. of apparent competition motifs (S4)	Number of apparent competition motifs, normalised by L/S	Meso	Milo <i>et al.</i> (2002); Stouffer <i>et al.</i> (2007)
No. of direct competition motifs (S5)	Number of direct competition motifs, normalised by L/S	Meso	Milo <i>et al.</i> (2002); Stouffer <i>et al.</i> (2007)

Metric	Definition	Scale	Reference
Generality	Standard deviation of normalised generality of all species within network, normalised by L/S	Micro	Williams & Martinez (2000)
Vulnerability	Standard deviation of normalised vulnerability of all species within network, normalised by L/S	Micro	Williams & Martinez (2000)

2.4 Extinction simulations and their evaluation

Following Dunhill *et al.* (2024), we simulated species loss from pre-extinction networks under trait-based, network-position-based, and random removal scenarios. Species were deleted sequentially, with cascading secondary extinctions allowed to propagate. Simulated post-extinction states were compared to observed (*i.e.*, reconstructed from fossil occurrence data) networks using mean absolute differences (MAD) of food web metrics Table 2 and modified true skill statistics (TSS) calculated separately at the node level (species presence/absence) and link level (presence/absence of interactions between species pairs). Scenarios were ranked within each reconstruction framework based on MAD and TSS performance, and Kendall’s rank correlation coefficient (τ) was used to quantify concordance in scenario ordering across reconstruction approaches. Detailed implementation of extinction sequences and secondary extinction rules is provided in Supplementary Material S1.

2.5 Software and Reproducibility

Ecological network reconstruction and extraction of structural metrics were conducted in Julia v1.11.4 (Bezanson *et al.*, 2017) using custom scripts and model implementations provided in Supplementary Material S1. All statistical analyses, including MANOVA, PERMANOVA, redundancy analysis, and post hoc comparisons, were performed in R

209 v4.5.2 (R Core Team, 2024). The full analytical workflow, including data preprocessing,
210 network generation, extinction simulations, and metric calculation, is fully reproducible
211 from the archived codebase.

212 The empirical data, derived network datasets, and all analysis scripts are archived in
213 a public Zenodo repository (**TODO** DOI), which includes a complete README file
214 describing software dependencies, execution order, and reproduction instructions.

215 **3 Results**

216 We found that different reconstruction approaches, even those that appeared structurally
217 similar, yielded fundamentally different ecological inferences. Across six reconstruction
218 approaches, inferred food web structure, species interactions, and extinction dynamics
219 differed consistently. Multivariate analyses showed pronounced separation among recon-
220 struction approaches in network metric space Figure 1, with reconstruction approach
221 explaining most of the variance in structural properties Figure 2. Notably, approaches
222 that were statistically similar in multivariate structural space often diverged in inferred
223 interactions Figure 3 or extinction dynamics Figure 4, demonstrating that structural
224 similarity does not guarantee concordance in species-level diets or trophic roles.

225 Reconstruction approach substantially influenced inferred extinction dynamics. Temporal
226 trajectories of network collapse, interaction loss, and motif reorganisation differed among
227 approaches Figure 2. Although node-level extinction rankings were often broadly consistent,
228 link-level outcomes and extinction inferences were highly sensitive to reconstruction
229 assumptions Figure 4. Together, these results show that ecological inferences drawn from
230 networks depend critically on the reconstruction framework used.

231 **3.1 Network structure differs among reconstruction approaches**

232 Across six reconstruction approaches, network structure (network properties listed in
233 Table 2) differed significantly (MANOVA, Pillai’s trace = 3.84, approximate $F_{40,11955}$
234 = 987.35, $p < 0.001$), indicating that the reconstruction approach systematically alters
235 inferred food web topology. Linear discriminant (LD) analysis identified two dominant axes
236 of variation, explaining 86% of the differences between reconstruction approach variance.
237 LD1 axis correlated with vulnerability, direct competition motifs, and connectance. LD2
238 axis correlated with maximum trophic level and apparent competition motifs, reflecting
239 vertical trophic structure (Figure 1; Table S1, Figure S1). Each of the all higher-order

240 variates explained less than 9% of the remaining variance.

241 [Figure 1 about here.]

242 **3.1.1 Variance partitioning of network structure**

243 Permutational multivariate analysis of variance (PERMANOVA) revealed that reconstruc-
244 tion framework accounted for the majority of variation in multivariate network structure
245 ($R^2 = 0.795$, $p < 0.001$), whereas temporal turnover across extinction phases explained a
246 comparatively small proportion of variance ($R^2 = 0.064$, $p < 0.001$). The reconstruction
247 approach $\times$ time interaction contributed a further 7.1% of variance ($R^2 = 0.071$, $p <$
248 0.001), indicating limited but significant time-dependent divergence among reconstruction
249 frameworks. Thus, differences among reconstruction approaches were more than an order of
250 magnitude greater than structural differences associated with ecological turnover through
251 the extinction sequence, even if the Pliensbachian–Toarcian dataset was characterised
252 with a significant community turnover.

253 To determine whether the dominance of the reconstruction framework reflected absolute
254 mean shifts among time bins, we repeated the analysis after centring network metrics
255 within each extinction phase. This procedure removes between-phase differences while
256 retaining within-phase structural variation. Even after temporal bin-standardised centring,
257 the reconstruction framework explained 84.8% of multivariate variance ($R^2 = 0.848$, $p <$
258 0.001). These results demonstrate that the influence of reconstruction approach is not
259 driven by temporal mean differences, but reflects intrinsic divergence among reconstruction
260 frameworks in how ecological interactions are organised.

261 **3.1.2 Statistical Drivers of Network Variation**

262 To identify which specific structural properties drive the multivariate separation observed
263 above, we partitioned variance at the level of individual network metrics. Results show
264 that reconstruction choice exerted a significantly stronger influence on network topology
265 than the ecological signal of species loss. In panel A of Figure 2 it is clear that for
266 certain network metrics reconstruction approaches predict different responses across
267 time *e.g.*, connectance, as well as differing magnitudes of change *e.g.*, the number of
268 apparent competition motifs. A two-way factorial ANOVA across all eight network metrics
269 confirmed that the reconstruction approach was the dominant driver of variance, with
270 partial eta-squared values (η_p^2) consistently exceeding 0.82 and reaching 0.97 for meso-scale

motifs (Figure 2, panel B; Table S3). While the extinction event (time bin) significantly altered network structure ($p < 0.001$), its relative importance remained secondary, typically explaining a smaller fraction of the total topological variation. This is clear in panel B of Figure 2 where all metrics are within the bottom-right (reconstruction approach-dominated) triangle (below the dashed line) of this space, emphasising that framework assumptions outweigh the ecological signal of species loss. Furthermore, the high inter-reconstruction approach Coefficient of Variation (CV) observed for some metrics (Table S4, Figure S4) highlights a sensitivity. The properties that are influenced by time are also those upon which the reconstruction approaches disagree most profoundly. Demonstrating that our understanding of structural food web collapse in the fossil record is highly contingent on the chosen reconstruction framework, particularly when examining complex trophic pathways beyond simple macro-scale properties like connectance.

[Figure 2 about here.]

3.1.3 Inferred pairwise interactions vary widely among reconstruction approaches

Despite some network reconstruction approaches showing similar network metrics, specific pairwise interactions often differed. Pairwise β_{OS} revealed that certain reconstruction approach pairs shared very few links Figure 3. Size-based models (ADBM, ATN) were broadly similar due to shared sole reliance on body-size constraints, whereas the Body-size ratio model exhibited consistently higher differences to other reconstruction approaches. PFWIM showed intermediate overlap with body mass-based models. These results demonstrate that agreement in overall network structure does not guarantee concordance in species-level interactions.

[Figure 3 about here.]

3.2 Reconstruction approach choice influences inferred extinction dynamics

To evaluate how reconstruction approach choice affects inferred extinction dynamics, we compared simulated post-extinction networks to observed networks using mean absolute differences (MAD) for network-level metrics and true skills statistics (TSS) for node- and link-level outcomes Figure 4. Across reconstruction approaches, MAD-based rankings were generally positively correlated (Kendall's $\tau = 0.13$ across structural metrics), indicating weak

302 but generally positive concordance on the relative importance of extinction drivers despite
303 substantial differences in reconstructed network structure. However, agreement within
304 the allometric models (ADMB, ATN) differed from patterns observed for reconstructed
305 network structure.

306 Node-level TSS rankings were similarly consistent across network reconstruction approaches
307 (Kendall's $\tau = 0.26 - 0.90$), reflecting broadly comparable node-level removal sequences.
308 In contrast, link-level outcomes were far more variable (Kendall's $\tau = -0.48 - 0.29$),
309 highlighting that inferences about which interactions are lost, retained, or re-established
310 during collapse and recovery are highly reconstruction approach contingent. Together,
311 these results suggest that while alternative reconstruction approaches converge on similar
312 species-level extinction patterns, the inferred pathways of interaction loss and cascading
313 dynamics depend strongly on both reconstruction approaches.

314 [Figure 4 about here.]

315 4 Discussion

316 4.1 Network reconstruction is not neutral: structural priors shape 317 ecological theory

318 Food web ecology has long treated network reconstruction as a methodologically neu-
319 tral step preceding ecological analysis, with limited attention to how methodological
320 choices might influence outcomes. Once a network is assembled (whether from obser-
321 vation, inference, or simulation) its properties are typically analysed as reflections of
322 underlying ecological organisation. Implicit in this workflow is a powerful assumption
323 that reconstructed networks provide structurally comparable representations of ecological
324 communities. As a result, differences in connectance, trophic structure, motif composition,
325 or robustness are interpreted as primarily reflecting biological variation, rather than the
326 effects of reconstruction approach.

327 Assuming that reconstructed food webs are independent of model choice is particularly
328 critical to evaluate within the context of deep-time palaeoecological data. Because
329 interactions in fossil ecosystems are never observed directly, and direct inferences (such
330 as gut contents) are limited, hence they must explicitly be formed through some form of
331 reconstruction approach. This necessity renders the underlying assumptions transparent
332 but also makes the resulting ecological narratives highly susceptible to the constraints

333 inherent in the chosen reconstruction framework. In these settings the risk is not just
334 incomplete data, but the potential for methodological artefacts to be misinterpreted as
335 genuine macroevolutionary or palaeoecological signals. Consequently, deep-time studies
336 offer a unique and stringent testing ground for determining whether community-level
337 responses (such as stability or collapse during mass extinctions) are robust features of the
338 ecosystem or merely byproducts of how we choose to construct the links between species.

339 Reconstruction framework explained the overwhelming majority of variance in inferred
340 food web topology, far outweighing the influence of temporal turnover across extinction
341 and recovery phases. Across an identical regional taxon pool, alternative reconstruction
342 approaches generated distinct structural signatures that occupied non-overlapping regions
343 of multivariate space Figure 1, demonstrating that divergence among reconstruction
344 approaches reflects intrinsic differences in how interactions are organised rather than
345 temporal shifts in community composition. Even after centring metrics within time bins
346 (extinction phases) to remove between-bin mean differences, reconstruction approach
347 remained the dominant driver of structural variation. These results indicate that recon-
348 struction approaches impose distinct ‘structural priors’ on ecological inference. These
349 priors are not subtle; they propagate into emergent topology, species roles, and predictions
350 of disturbance dynamics. Network structure is therefore not solely a property of ecological
351 communities, but jointly determined by ecological data and assumptions made when
352 inferring/reconstructing interactions (Gauzens *et al.*, 2025; Strydom *et al.*, 2026).

353 Crucially, this dominance was not confined to multivariate summaries. Variance partition-
354 ing at the level of individual network properties revealed that reconstruction approach
355 overwhelmingly structured specific ecological metrics, thus the imprint of the recon-
356 struction framework is visible not only in aggregate topology but in the very structural
357 features often interpreted as ecological signals. Notably, the few properties that exhibited
358 detectable temporal sensitivity were also those with the greatest inter-reconstruction
359 approach disagreement (*i.e.*, motif distributions), indicating that temporal trends are
360 most difficult to disentangle when reconstruction frameworks diverge most strongly. These
361 results suggest that structural assumptions do not merely shift networks within a shared
362 architectural space; they condition the specific patterns through which ecological change
363 is perceived and interpreted.

364 This has direct implications for the interpretation of comparative network studies. Feasible,
365 realised, and structural reconstruction approaches encode different assumptions about

constraint, optimisation, and topology, with these assumptions propagating into emergent metrics and dynamical predictions (Dunne *et al.*, 2002; Curtsdotter *et al.*, 2011; Allesina & Tang, 2012; Poisot & Gravel, 2014; Michalska-Smith & Allesina, 2019). When networks reconstructed under different classes are compared across spatial gradients, disturbance regimes, or evolutionary transitions, part of the observed variation may derive from reconstruction choice rather than ecological process. Without explicit standardisation or sensitivity analysis, methodological heterogeneity can be mistaken for biological signal. Food web ecology has devoted substantial effort to understanding how topology shapes dynamics; comparatively less attention has been paid to how reconstruction method shapes topology. Our findings indicate that these two questions cannot be separated.

4.2 Scale-dependent robustness in network-based inference

Importantly, reconstruction sensitivity was not uniform across levels of inference. Node-level extinction rankings were broadly consistent among reconstruction approaches, whereas interaction-level outcomes and cascade trajectories were highly contingent on reconstruction methods. The predominance of the reconstruction framework over temporal turnover helps explain this pattern. Different reconstruction approaches often converged on similar patterns of community vulnerability, yet diverged substantially in the mechanisms through which collapse unfolded. Broad ecological patterns can emerge across multiple plausible interaction architectures, whereas conclusions about which interactions are lost, retained, or reorganised depend strongly on how those interactions are inferred.

This distinction challenges a central ambition of food web ecology: using interaction structure to identify the mechanisms underlying stability and collapse. Our findings suggest that while broad patterns may be robust across reconstruction approaches, mechanistic explanations are far less secure. Had Dunhill *et al.* (2024) used a reconstruction approach other than PFWIM, the inferred drivers of extinction and recovery may have differed substantially. If cascade pathways vary across equally plausible reconstructions, then mechanistic narratives derived from a single inferred topology may overstate their precision (Dunne *et al.*, 2002; Curtsdotter *et al.*, 2011; Allesina & Tang, 2012). The apparent determinism of extinction cascades may therefore reflect reconstruction-imposed structure as much as ecological inevitability.

For macroecology, this metric dependence clarifies where network-based inference is accurate. Aggregate properties may be comparatively robust to reconstruction assumptions,

whereas conclusions about interaction turnover, motif reorganisation, or fine-scale trophic dynamics are intrinsically uncertain. Recognising this asymmetry is essential if network analyses are to inform comparative synthesis across space and time.

Taken together, these results underscore that network reconstruction is not a neutral preprocessing step but an additional part of the hypothesis-generating process in which each reconstruction approach encodes a distinct set of ecological assumptions. The inferred topology and dynamics of a food web therefore reflect not only ecological data, but the theoretical assumptions embedded in the reconstruction framework. Disagreement among reconstruction approaches does not imply that any single approach is ‘wrong’, but rather that different reconstruction approaches capture different facets of ecological reality (Petchey *et al.*, 2011; Stouffer, 2019). Disagreement among reconstruction approaches does not imply that any single approach is ‘incorrect’. Rather, different reconstruction approaches capture different facets of ecological constraints (*e.g.*, trait compatibility, energetic optimisation, or topological regularity) and thus is a case of matching the ‘correct’ reconstruction approach to the task at hand.

This perspective reframes reconstruction choice as part of hypothesis specification. Researchers must align reconstruction approaches with the ecological signals of interest (whether potential interactions, realised diets, or macro-scale structural properties) rather than treating network reconstruction as a technical convenience. Viewed through the lens of accuracy and precision, our results indicate that some network-based inferences are relatively robust across reconstruction approaches, whereas others remain intrinsically uncertain. High-level extinction rankings were broadly convergent, suggesting relative accuracy at coarse resolution, but interaction-level details and temporal cascade dynamics diverged substantially, indicating limited precision in reconstructing the fine structure of collapse. Recognising and explicitly accounting for this distinction is essential if food web ecology is to move beyond descriptive reconstruction toward rigorous comparative inference.

4.3 Implications for comparative biogeography and global change research

Network approaches are increasingly applied to examine how ecological organisation varies across latitudinal gradients, environmental filters, disturbance regimes, and climate-driven transitions (Tylianakis *et al.*, 2008; Gilman *et al.*, 2010). In global change ecology, networks

are used to project vulnerability under warming, quantify rewiring of interactions, and assess stability under species loss (*e.g.*, Hao *et al.*, 2025; Marjakangas *et al.*, 2025). These studies frequently interpret variation in connectance, trophic height, interaction β -diversity, or robustness as indicators of ecological differentiation among regions or time intervals (*e.g.*, Pellissier *et al.*, 2018; Trøjelsgaard & Olesen, 2016). Our results show that reconstruction choice can systematically alter inferred topology and disturbance dynamics even when species composition is held constant. This suggests that apparent differences in network structure across spatial or climate gradients may reflect variation in reconstruction as much as ecological process.

Deep-time palaeo food webs provide a complementary perspective because they capture ecosystem responses to large-scale environmental perturbations and extinction events under past climate change (*e.g.*, Dunhill *et al.* (2024); Smith *et al.* (2025); Karapınar *et al.* (2026)). Fossil networks therefore represent natural experiments for evaluating resilience, trophic reorganisation, and recovery following extreme environmental change. Studies of palaeo food webs have demonstrated how network structure mediates extinction cascades and post-disturbance reassembly (Roopnarine, 2006; Dunne *et al.*, 2008), providing empirical constraints on long-term ecological stability.

However, our results emphasise that even in deep-time systems structural conclusions remain sensitive to reconstruction approaches. Treating reconstructed networks as reconstruction dependent inferences rather than single deterministic representations provides a more transparent framework for incorporating uncertainty into comparative macroecology and for using palaeo data to inform expectations about modern climate-driven reorganisation.

4.4 Toward a more explicit modelling paradigm in food web ecology

The broader implication is not that any single reconstruction framework is ‘correct’ or ‘incorrect’. Rather, each reconstruction approach represents a distinct hypothesis about how interactions are constrained by trait compatibility, energetic optimisation, or topological regularity (Petchey *et al.*, 2011). Food web reconstruction is therefore theory-laden. Making this explicit shifts reconstruction from a preparatory step to a central component of ecological modelling.

A mature modelling paradigm in food web ecology would treat reconstruction approaches

as testable, incorporate probabilistic link inference where possible, and quantify the sensitivity of macroecological conclusions to alternative representations of interaction structure. Such an approach aligns with recent advances in probabilistic and ensemble network modelling and would strengthen the interpretability of network-based inference under global change (Poisot *et al.*, 2016; Banville *et al.*, 2025).

5 Conclusions

Ecological network reconstruction is not a neutral technical procedure but a theoretical act that shapes ecological inference. By applying six contrasting reconstruction frameworks to an identical species pool, we show that different reconstruction approaches systematically influence inferred food web topology, interaction identity, and disturbance dynamics. Some coarse-grained patterns, such as relative species vulnerability, are comparatively robust across representations. In contrast, fine-scale interaction structure and cascade pathways are highly contingent on reconstruction approaches. The reliability of network-based inference is therefore scale dependent.

These results challenge the implicit assumption that reconstructed networks are comparable across systems — whether comparing modern communities across environmental gradients or fossil assemblages across extinction intervals. When reconstruction frameworks differ, variation in connectance, trophic organisation, robustness, or interaction turnover may reflect embedded reconstruction approaches as much as ecological processes. Network reconstruction should thus be treated as an explicit component of hypothesis specification in comparative macroecology and biogeography.

No single reconstruction approach captures the full complexity of ecological organisation, but neither are alternative reconstruction approaches interchangeable. Aligning reconstruction framework with inferential goals, standardising approaches across comparative studies, and incorporating ensemble or probabilistic representations will be essential for strengthening the interpretability of network analyses across spatial and temporal gradients, including efforts to use deep-time systems to inform expectations under contemporary climate change. As ecological networks increasingly inform global change research, recognising network reconstruction as fundamental determinants of inference is critical for advancing food web ecology from descriptive reconstruction toward rigorous comparative synthesis.

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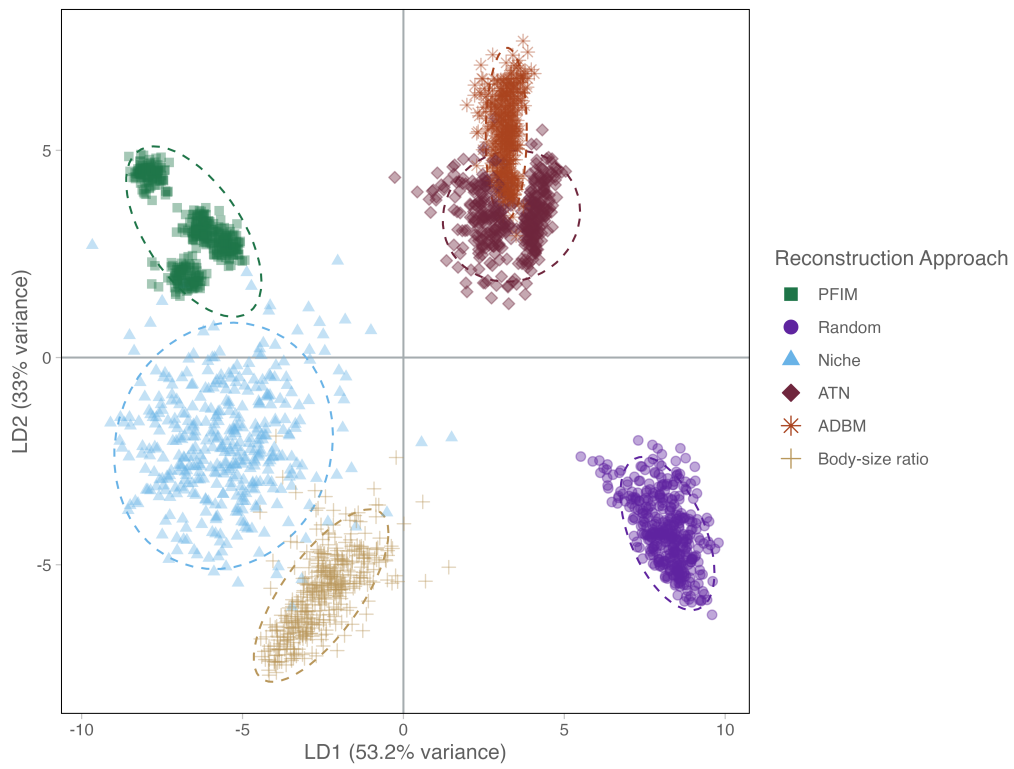


Figure 1: Linear discriminant analysis (LDA) of ecological network metrics for six reconstruction approach types. Each point represents a replicate, and ellipses indicate 95% confidence regions for each reconstruction approach.

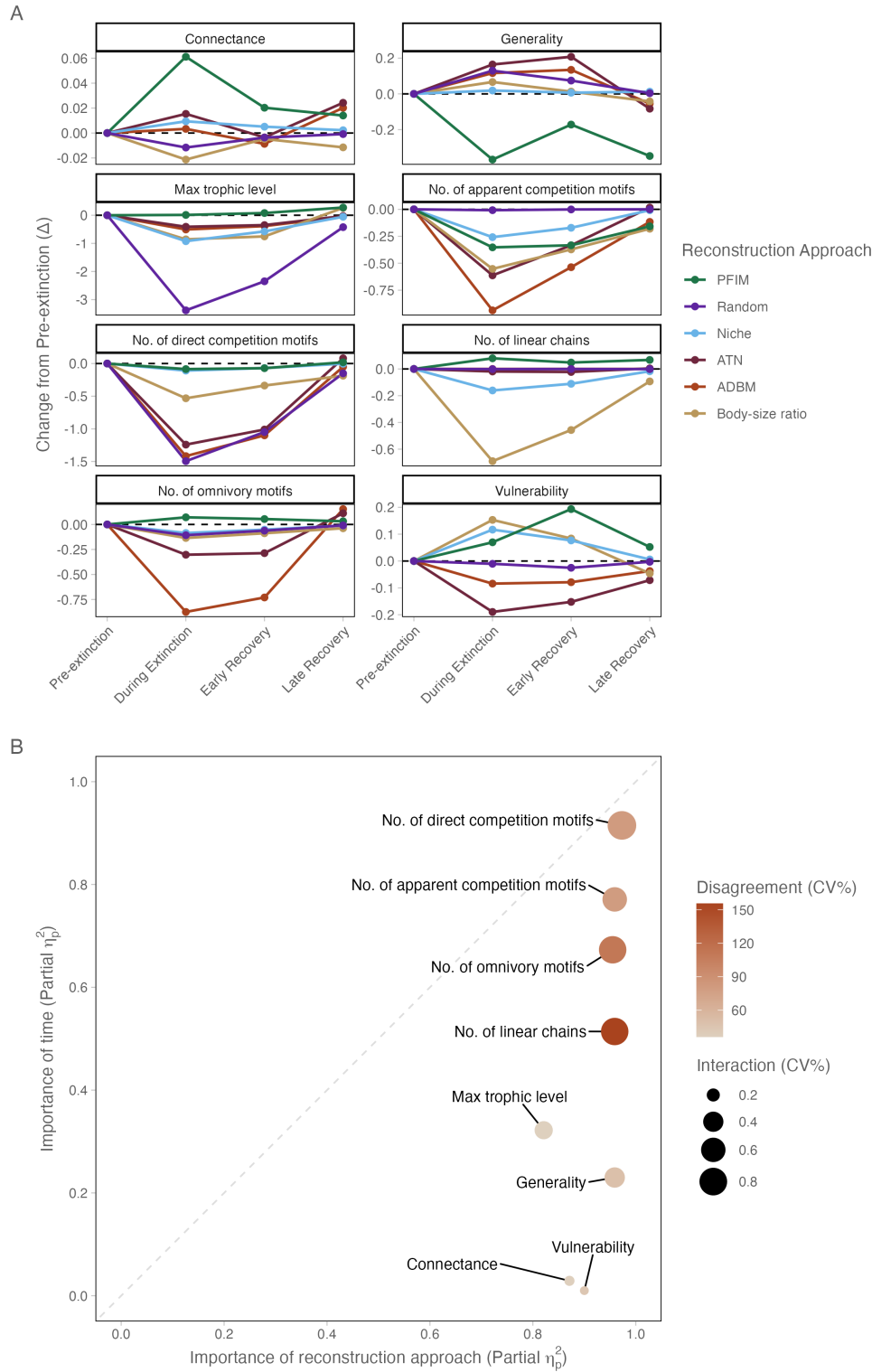


Figure 2: Figure 2: Temporal changes in food-web structure and the relative influence of reconstruction approach and ecological turnover on network properties. **(A)** Changes in network metrics through time, expressed as deviations from pre-extinction conditions (δ). Metrics are faceted and scaled independently to highlight differences in temporal trajectories among network properties. **(B)** Relative influence of reconstruction approach (x-axis) and time (y-axis) on individual network metrics. The dashed 1:1 line indicates equal explanatory power. Metrics below the line are more strongly influenced by reconstruction approach than by temporal turnover. Bubble size represents the reconstruction approach $\times$ time interaction, and colour indicates disagreement among reconstruction approaches (mean CV%), with darker colours indicating greater divergence in metric estimates.

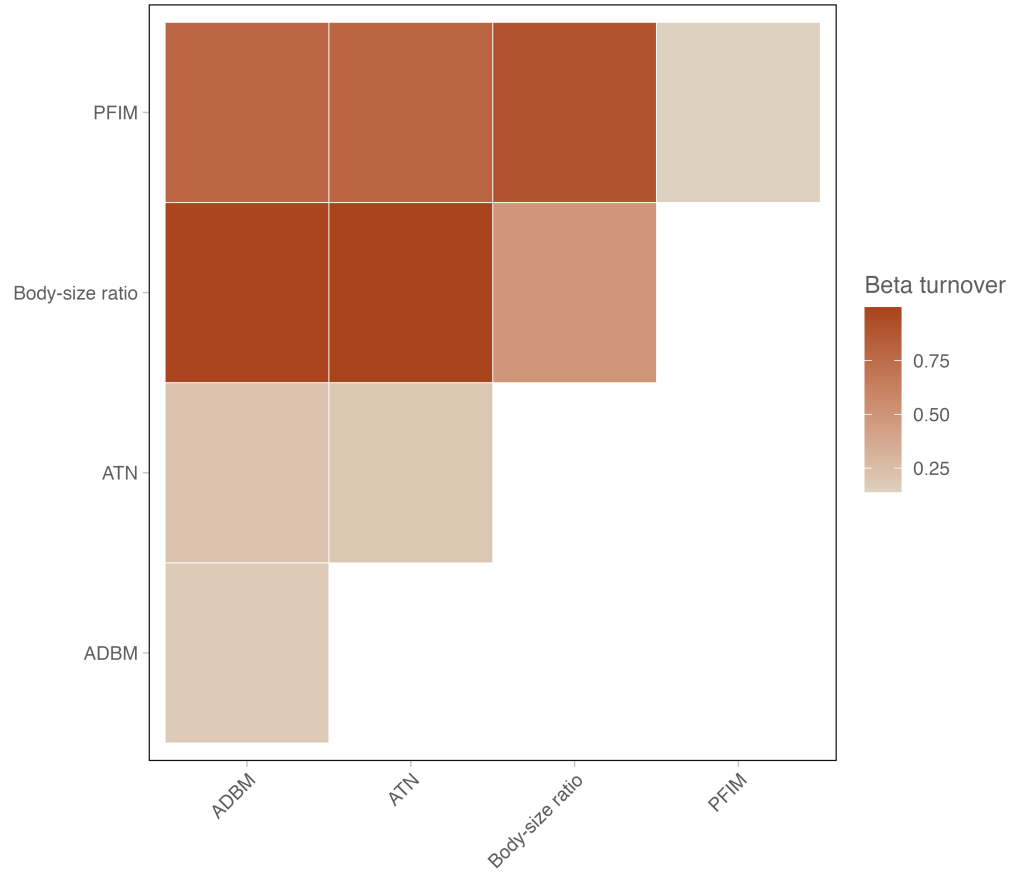


Figure 3: Pairwise interaction turnover (β_{OS}) among four food-web reconstruction approaches that infer species-specific trophic interactions (ADBM, ATN, Body-size Ratio, and PFWIM). Each cell represents the mean turnover value between a pair of reconstruction approaches, with darker colours indicating greater dissimilarity in inferred interactions. High turnover values indicate strong disagreement in predicted trophic links, whereas lower values indicate greater similarity.

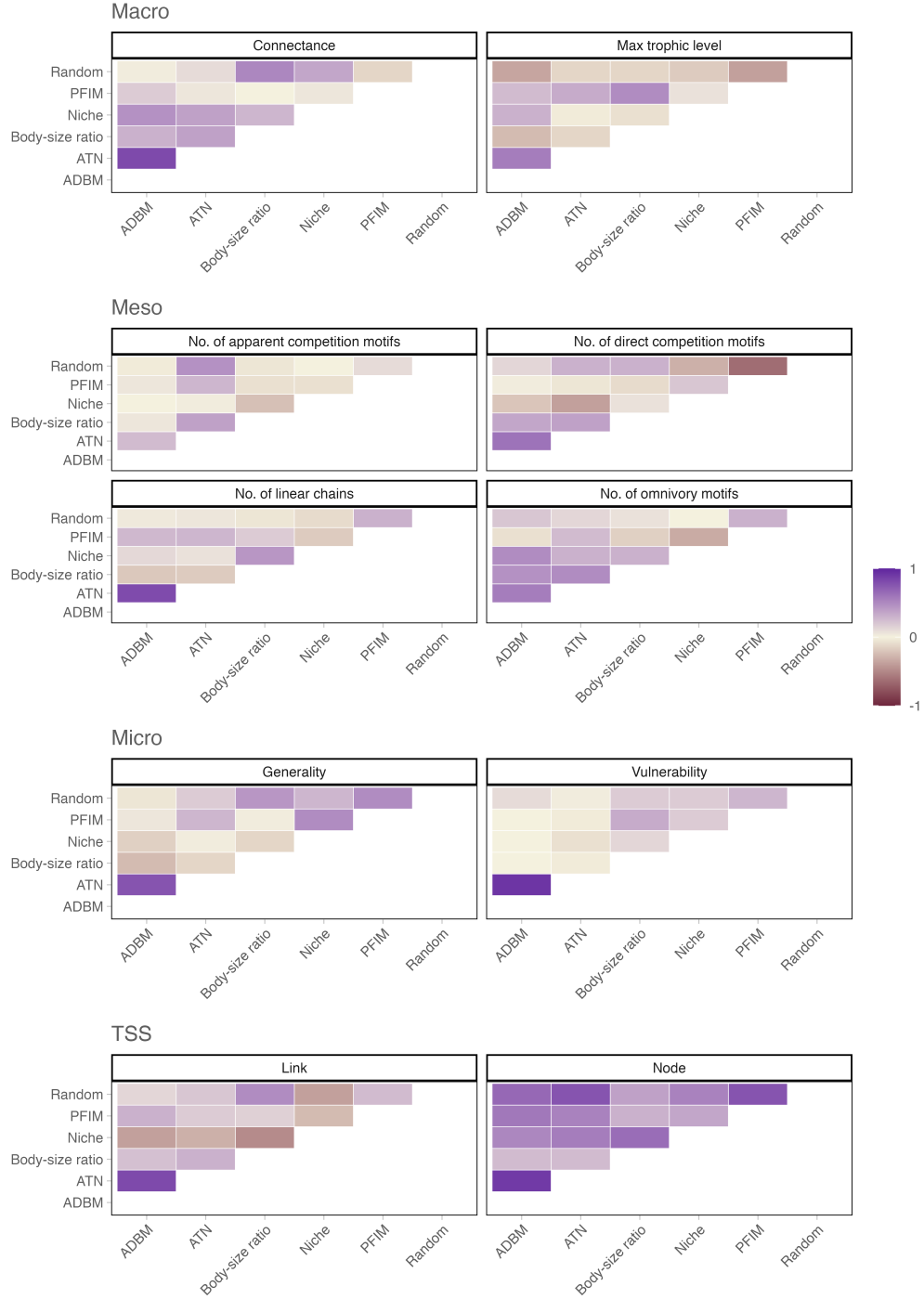


Figure 4: Heatmaps showing pairwise Kendall rank correlation coefficients (τ) between reconstruction approaches for each network metric. Each panel corresponds to a different metric and displays the degree of agreement in extinction-scenario rankings across reconstruction approaches based on mean absolute differences (MAD) between observed and predicted network values as well as differences in species links (TSS). Positive τ values (purple) indicate concordant rankings between reconstruction approaches, whereas negative τ values (sienna) indicate opposing rankings. Paler colours approaching zero represent little or no agreement.